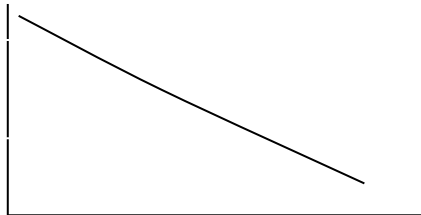


1(a)	<i>independent variable:</i> <u>concentration</u> of, copper (II) sulfate ; <i>dependent variable:</i> <u>volume</u> of, oxygen ;	2
1(b)(i)	1 minimum of five <u>stated</u> dilutions <u>and</u> correct units ; 2 method of <u>serial</u> dilution for a minimum of 2 intermediate (concentrations) ;	2

Question	Answer	Marks																	
1(b)(ii)	any six from: - maintaining a constant temperature and a suitable method ; - maintaining a constant pH <u>and</u> use a buffer ; <i>mp 3 and 4 and 5: if 'amount' rather than 'volume', penalise only once and ecf the following ones</i> - same / stated, volume of catalase / enzyme or same / stated, concentration of catalase / enzyme ; - same / stated, volume of each copper sulfate / inhibitor solution ; - measure / record, volume of oxygen / gas collected, in set time / set intervals or time taken to reach a set volume ; - state a suitable control ; - use new / fresh (named) solutions for each replicate / experiment ; - repeat at least twice / three replicates, <u>and</u> finding mean for at least one concentration ; - named hazard <u>and</u> risk <u>and</u> precaution ; e.g. <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td rowspan="2">catalase / enzyme</td><td>irritant / allergy</td><td rowspan="8">wear gloves / goggles / mask PPA</td></tr> <tr> <td>moderate / medium, hazard</td></tr> <tr> <td rowspan="3">hydrogen peroxide</td><td>irritant / allergy</td></tr> <tr> <td>moderate / medium, hazard</td></tr> <tr> <td>oxidising agent</td></tr> <tr> <td rowspan="4">copper(II) sulfate</td><td>irritant / allergy</td></tr> <tr> <td>moderate / medium, hazard</td></tr> <tr> <td>corrosive</td></tr> <tr> <td>toxic</td><td>do not pour down sink</td></tr> </tbody> </table> 10 AVP ;	hazard	risk	precaution	catalase / enzyme	irritant / allergy	wear gloves / goggles / mask PPA	moderate / medium, hazard	hydrogen peroxide	irritant / allergy	moderate / medium, hazard	oxidising agent	copper(II) sulfate	irritant / allergy	moderate / medium, hazard	corrosive	toxic	do not pour down sink	6
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1(b)(iii)	1 axes correctly orientated with labels ; 2 axes labels have correct units ; 3 line shows downward trend as concentration of copper(II) sulfate increases ; e.g. volume of oxygen / mm³ or cm³  concentration of copper sulfate / mol dm⁻³	3
1(c)(i)	All three lines correctly drawn / indicated ; $V_{\max}$ or $\frac{1}{2} V_{\max}$ correctly labelled for lines drawn ; K_m correctly labelled (on correct line) ;	3
1(c)(ii)	plateaus below $V_{\max}$ / does not reach $V_{\max}$ / $V_{\max}$ is lower (than without inhibitor) ; same / almost same / similar, K_m ;	2

Question	Answer	Marks
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